Pratyush Bhattacharya

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Professional Summary

B.S. Computer Science candidate at UC San Diego and current software engineering intern, shipping production TypeScript and React Native features on a Dockerized, AWS-deployed stack. Experienced across the full stack, spanning RESTful API design, SQL data modeling, third-party OAuth integrations, and LLM-backed application workflows, on a strong foundation in data structures, algorithms, and object-oriented design. Proficient in TypeScript, Python, JavaScript, Java, and C++. Proven contributor in Agile team environments through sprint planning, pair programming, and iterative code reviews.

Experience

Lyntra, Inc. Aug 2026 - Present
Software Engineering Intern | TypeScript, React, React Native, Node.js, Docker, AWS

- Diagnosed five parallel design systems in a 117k-LOC React Native app and replaced them with one enforced layer of 26 shared primitives, cutting raw hex color literals 51% (1,512→735) and hard-coded font sizes 47% (1,271→674).
- Removed 19,071 lines across 50 files by measuring static reachability before cutting rather than inferring from screen names; one pass deleted 11,701 lines across 31 files after analysis showed that severing four surfaces cascaded to 26 files, not the ~3,700 lines estimated from route files alone.
- Design and implement server-side REST endpoints and OAuth 2.0 integrations, validated with Postman collections covering authentication, error, and empty-state contracts, with secure handling of student academic records.
- Implement LLM workflows for task decomposition, prioritization, retrieval-augmsented guidance, tool calling, and agent-style orchestration, all behind a data-source seam that keeps model-dependent flows testable offline.

Education

University of California San Diego Expected Jun 2027
B.S. in Computer Science
Relevant Coursework: Advanced Data Structures, Software Engineering, Machine Learning, Deep Learning, Server-side Analytics, Basic Cybersecurity

CodeSignal Sep 2025
Full-Stack Engineering Certification
Relevant Coursework: React.js, Node.js, Express.js, HTML, CSS

Trinity College London Nov 2021
Acting Certification
Relevant Coursework: Communication, Public Speaking, Improvisation

Projects

Website for Data Collection Jan 2026 - Mar 2026
Solo Developer | JavaScript, Node.js, PHP, SQL, HTML/CSS

- Built a full-stack web application as sole developer, designing and deploying responsive HTML/CSS/JavaScript interfaces while configuring Apache virtual hosts and Node.js servers to support dynamic web services and analytics collection.
- Developed RESTful APIs and end-to-end data pipelines using Node.js, HTTP, and SQL databases, implementing secure server-side validation, session management, and data reporting dashboards to transform user interactions into actionable analytics.
- Applied modern web architecture principles including client-server design, MVC patterns, and state management; leveraged AI-assisted development tools to accelerate front-end prototyping and data visualization workflows.

Java Chess Game Jul 2025 - Aug 2025
Developer | Java | github.com/PratyushBhattacharya1/ChessGame

- Architected and implemented a feature-complete chess engine in Java, covering move generation, rule enforcement, and user interaction.
- Designed custom algorithms for move validation and board visualization, selecting optimized data structures to enhance robustness.
- Applied object-oriented design principles and key design patterns (Dependency Inversion, Context) to modularize components and streamline future feature expansion.

Routines App Jan 2024 - Mar 2024
Programmer | Java, SQLite, Android Studio, JUnit

- Collaborated with a six-person team in an Agile workflow to build an Android routines app from concept to deployment.
- Developed backend services with a relational SQL database and crafted responsive Java-based UIs in Android Studio.
- Drove code quality through JUnit test coverage, pair programming, GitHub version control, and iterative code reviews, delivering a stable release on schedule.

Activities & Leadership

Lyntra, Inc Aug 2026 - Present
Software Engineer

- Authored a 1,874-line remediation plan and companion tracker decomposing remaining frontend work into 56 batches across 12 dependency-ordered sets — each with owner, effort estimate, blocking relationships and an exit-gate evidence table — plus an 8-step per-batch workflow, standing in for an issue tracker the team did not have.

International School of Choueifat, Dubai Sep 2022 - Mar 2023

Head of Academics

- Led academic strategy and performance initiatives across all high school grades, spearheading school-wide engagement through targeted, student-led interventions.
- Designed and launched a peer mentorship program pairing struggling students with top performers, and organized grade-level teams to build shared quiz banks and collective study resources.

SKILLS

Programming Languages: Python, TypeScript, JavaScript, Java, C++, C, SQL, HTML/CSS

Frameworks & Libraries: React, React Native, Node.js, Express.js, scikit-learn, NumPy, Pandas

Databases: PostgreSQL, SQLite

Tools & Platforms: Git/GitHub, GitHub Actions (CI/CD), Docker, Postman, ESLint, JUnit, Android Studio, Figma, Apache, Claude Code

Technical Strengths: Data Structures, Algorithms, Object-Oriented Design (SOLID), RESTful API Design, OAuth 2.0, Design Systems, Automated Testing, Agile/Scrum, Code Review, LLM Integration, Retrieval-Augmented Generation (RAG), Tool Calling, Machine Learning, Deep Learning, Performance Optimization

Spoken Languages: English (Native), French (Conversational)